

Functions of Two or Three Real Variables (2005–2014)

Category: Multivariable Calculus and Differential Equations • Official Mock Test Series

TOTAL QUESTIONS

11

TOTAL MARKS

48 M

TEST DURATION

33 Min

PAPER PATTERN

Classic Paper Pattern (2

Section / Type		Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)		11	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2014 • Q25 • ID: JAM_2014_Q25)

MCQ

+2 M

Neg: -0.67

Q.25 For all $(x, y) \in \mathbb{R}^2$, let $f(x, y) = \begin{cases} \frac{x}{|x|} \sqrt{x^2 + y^2} & \text{if } x \neq 0, \\ 0 & \text{if } x = 0. \end{cases}$

Then $\frac{\partial f}{\partial x}(0, 0) + \frac{\partial f}{\partial y}(0, 0)$ equals

- (A) -1 (B) 0 (C) 1 (D) 2

Question 2 (JAM 2014 • Q24 • ID: JAM_2014_Q24)

MCQ

+2 M

Neg: -0.67

Q.24 For all $(x, y) \in \mathbb{R}^2$, let $f(x, y) = \begin{cases} x & \text{if } y = 0, \\ x - y^3 \sin(1/y) & \text{if } y \neq 0. \end{cases}$

Then at the point $(0, 0)$,

- (A) f is NOT continuous
(B) f is continuous but NOT differentiable
(C) $\frac{\partial f}{\partial x}$ exists but $\frac{\partial f}{\partial y}$ does NOT exist
(D) f is differentiable

Question 3 (JAM 2014 • Q5 • ID: JAM_2014_Q5)

MCQ

+1 M

Neg: -0.33

Q.5 If $f(x, y, z) = x^2y + y^2z + z^2x$ for all $(x, y, z) \in \mathbb{R}^3$ and $\nabla = \frac{\partial}{\partial x}\mathbf{i} + \frac{\partial}{\partial y}\mathbf{j} + \frac{\partial}{\partial z}\mathbf{k}$, then the value of $\nabla \cdot (\nabla \times \nabla f) + \nabla \cdot (\nabla f)$ at $(1, 1, 1)$ is

- (A) 0 (B) 3 (C) 6 (D) 9

Question 4 (JAM 2014 • Q3 • ID: JAM_2014_Q3)

MCQ

+1 M

Neg: -0.33

Q.3 Let $f(x, y) = \sum_{k=1}^{10} (x^2 - y^2)^k$ for all $(x, y) \in \mathbb{R}^2$. Then for all $(x, y) \in \mathbb{R}^2$,

(A) $x \frac{\partial f}{\partial x}(x, y) - y \frac{\partial f}{\partial y}(x, y) = 0$

(B) $x \frac{\partial f}{\partial x}(x, y) + y \frac{\partial f}{\partial y}(x, y) = 0$

(C) $y \frac{\partial f}{\partial x}(x, y) - x \frac{\partial f}{\partial y}(x, y) = 0$

(D) $y \frac{\partial f}{\partial x}(x, y) + x \frac{\partial f}{\partial y}(x, y) = 0$

Question 5 (JAM 2012 • Q7 • ID: JAM_2012_Q7)

MCQ

+6 M

Neg: -2.0

Q.7 For $c > 0$, if $a\hat{i} + b\hat{j} + c\hat{k}$ is the unit normal vector at $(1, 1, \sqrt{2})$ to the cone $z = \sqrt{x^2 + y^2}$, then

(A) $a^2 + b^2 - c^2 = 0$

(B) $a^2 - b^2 + c^2 = 0$

(C) $-a^2 + b^2 + c^2 = 0$

(D) $a^2 + b^2 + c^2 = 0$

Question 6 (JAM 2012 • Q3 • ID: JAM_2012_Q3)

MCQ

+6 M

Neg: -2.0

Q.3 If $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ is defined by

$$f(x, y) = \begin{cases} \frac{x^3}{x^2 + y^4} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0), \end{cases}$$

then

(A) $f_x(0, 0) = 0$ and $f_y(0, 0) = 0$

(B) $f_x(0, 0) = 1$ and $f_y(0, 0) = 0$

(C) $f_x(0, 0) = 0$ and $f_y(0, 0) = 1$

(D) $f_x(0, 0) = 1$ and $f_y(0, 0) = 1$

Question 7 (JAM 2009 • Q11 • ID: JAM_2009_Q11)

MCQ

+6 M

Neg: -2.0

Q.11 Let $T(x, y, z) = xy^2 + 2z - x^2z^2$ be the temperature at the point (x, y, z) . The unit vector in the direction in which the temperature decreases most rapidly at $(1, 0, -1)$ is

(A) $-\frac{1}{\sqrt{5}}\hat{i} + \frac{2}{\sqrt{5}}\hat{k}$

(B) $\frac{1}{\sqrt{5}}\hat{i} - \frac{2}{\sqrt{5}}\hat{k}$

(C) $\frac{2}{\sqrt{14}}\hat{i} + \frac{3}{\sqrt{14}}\hat{j} + \frac{1}{\sqrt{14}}\hat{k}$

(D) $-\left(\frac{2}{\sqrt{14}}\hat{i} + \frac{3}{\sqrt{14}}\hat{j} + \frac{1}{\sqrt{14}}\hat{k}\right)$

Question 8 (JAM 2008 • Q3 • ID: JAM_2008_Q3)

MCQ

+6 M

Neg: -2.0

Q.3 Let $s_n = \int_0^1 \frac{nx^{n-1}}{(1+x)} dx$ for $n \geq 1$. Then as $n \rightarrow \infty$, the sequence $\{s_n\}$ tends to

(A) 0

(B) $1/2$

(C) 1

(D) $+\infty$

Question 9 (JAM 2007 • Q9 • ID: JAM_2007_Q9)

MCQ

+6 M

Neg: -2.0

9. Let $f: \mathbf{R}^2 \rightarrow \mathbf{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{x^2y}{x^4 + y^2} & \text{if } (x, y) \neq (0, 0), \\ 0 & \text{if } (x, y) = (0, 0). \end{cases}$$

Which of the following statements holds regarding the continuity and the existence of partial derivatives of f at $(0, 0)$?

(A) Both partial derivatives of f exist at $(0, 0)$ and f is continuous at $(0, 0)$

(B) Both partial derivatives of f exist at $(0, 0)$ and f is NOT continuous at $(0, 0)$

(C) One partial derivative of f does NOT exist at $(0, 0)$ and f is continuous at $(0, 0)$

(D) One partial derivative of f does NOT exist at $(0, 0)$ and f is NOT continuous at $(0, 0)$

Question 10 (JAM 2006 • Q3 • ID: JAM_2006_Q3)

MCQ

+6 M

Neg: -2.0

3. Let $f(x, y) = x^5 y^2 \tan^{-1}\left(\frac{y}{x}\right)$. Then $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$ equals

(A) $2f$

(B) $3f$

(C) $5f$

(D) $7f$

Question 11 (JAM 2005 • Q11 • ID: JAM_2005_Q11)

MCQ

+6 M

Neg: -2.0

11. Let $f : \mathbf{R} \longrightarrow \mathbf{R}$ be continuous and $g, h : \mathbf{R}^2 \longrightarrow \mathbf{R}$ be differentiable. Let $F(u, v) = \int_v^u f(t) dt$,

where $u = g(x, y)$ and $v = h(x, y)$. Then $\frac{\partial F}{\partial x} + \frac{\partial F}{\partial y} =$

(A) $f(g(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] - f(h(x, y)) \left[\frac{\partial h}{\partial x} + \frac{\partial h}{\partial y} \right]$

(B) $f(h(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] - f(g(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$

(C) $f(h(x, y)) \left[\frac{\partial g}{\partial x} + \frac{\partial g}{\partial y} \right] + f(g(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$

(D) $f(g(x, y)) \left[\frac{\partial g}{\partial x} - \frac{\partial g}{\partial y} \right] + f(h(x, y)) \left[\frac{\partial h}{\partial x} - \frac{\partial h}{\partial y} \right]$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

Functions of Two or Three Real Variables (2005–2014) • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+2	C
2	MCQ	+2	D
3	MCQ	+1	C
4	MCQ	+1	D
5	MCQ	+6	A
6	MCQ	+6	B

Q. No.	Type	Marks	Official Key
7	MCQ	+6	B
8	MCQ	+6	B
9	MCQ	+6	B
10	MCQ	+6	D
11	MCQ	+6	A

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt ($-1/3$ for 1-mark questions, $-2/3$ for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., $[a, b]$) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.